High-Throughput Portable True Random Number Generator Based on Jitter-Latch Structure

Xinyu Wang, Huaguo Liang, Yanjie Wang, Liang Yao, Yang Guo, Maoxiang Yi, *Associate Member, IEEE*, Zhengfeng Huang, Haochen Qi, and Yingchun Lu

***Abstract*— Under the requirement of highly reliable encryption, the design of true random number generators (TRNGs) based on field-programmable gate arrays (FPGAs) is receiving increased attention. Although TRNGs based on ring oscillators (ROs) and phase-locked loops (PLLs) have the advantages of small resource overhead and high throughput, there are problems such as instability of randomness and poor portability. To improve the randomness, portability, and throughput of a random number generator, we design a TRNG whose randomness is generated by the oscillation of self-timed rings (STRs) and accurately extracted by a jitter-latch structure. The portability of the structure is verified by electronic design automation (EDA) tools. Under the condition of 0°C–80°C ambient temperature and 1.0 ± 0.1 V output voltage, the proposed structure is tested many times on Xilinx Spartan-6 and Virtex-6 FPGAs with an automatic routing mode. Theoretical analysis shows that this method can effectively improve the coverage of jitter and reduce the migration phenomenon. Experimental results show excellent performance in randomness, robustness, and portability, and the throughput reaches 100 Mbps.**

***Index Terms*— TRNG, FPGA, STRs, jitter-latch structure, portability.**

I. Introduction

IN THE era of information explosion, [1] information security issues are of broad concern. Random numbers used as ciphers, public keys, initialization vectors, and filling values are crucial to security [2]. In a highly reliable information encryption environment, true random number generators (TRNGs) based on field-programmable gate arrays (FPGAs) and other hardware perform better than those based on software, and have attracted more attention [3]–[5].

The circuit structure used to generate random phenomena in TRNGs is called an entropy source. Commonly used entropy sources in FPGAs are Phase locked loops (PLLs), [6], [7] digital clock management (DCM), [8], [9] ring oscillators (ROs), [10]–[14] and self-timed rings (STRs). References [15]–[18] Petra *et al.* designed a TRNG using PLL as the entropy source circuit [6]; it shows good randomness, but its portability is limited for the PLL design, and it must be readjusted often to ensure a highly random output. Similar problems are observed in entropy source circuits based on ROs. Strict path delay is required in TRNGs. A TRNG based on a programmable delay line (PDL) was proposed by Anandakumar *et al.*, [14] and Rozic *et al.* proposed one based on a fast carry chain, [11] with the intent to reduce the delay caused by process deviation. These require complicated placing and routing steps to increase the randomness. The portability of these structures was limited, which led to the dependence of randomness on the degree of delay adjustment.

Considering that STRs are more robust than ROs, [19] many scholars have conducted research on them as entropy sources of TRNGs. Cherkaoui *et al.* used an STR instead of an RO to design a true random number generator, but the design consumed many resources and much power [15]. Martin *et al.* used coherent sampling STRs to extract true random numbers, [16] which reduced the overhead of TRNG while ensuring randomness, but its throughput rate was low. Zhang *et al.* used fast forward on FPGAs to capture the signal edge positions of STR loops as logic primitives [18]. However, this method required manual placement and lost the portability of the structure. Liu *et al.* proposed a staged-running TRNG structure, [17] which used the unstable state in STRs. The piecewise running strategy adopted in the design affected the throughput of the generator, and the randomness quality was not ideal. However, they proposed the idea of quantifying jitter as unstable data bits in the process of state transition. Inspired by the design of Liu *et al.*, this paper proposes a TRNG based on a jitter-latch structure to improve the portability and the randomness difference between devices. By designing the Muller C-element of the look-up-table (LUT), the uncertain stable state caused by jitter in STRs is sampled and latched, so as to accurately sample jitter. This paper makes the following contributions: 1) A jitter-latch structure is designed, which increases the output entropy of the structure by increasing the sampling coverage of jitter. 2) A post-processing structure is designed to filter out most errors in automatic routing. The designed TRNG has a throughput of 100 Mbps and can be realized by automatic routing. Experimental results on Xilinx FPGAs show that its output has good randomness and stability.

The remainder of this paper is structured as follows. Section II introduces the generation principle of randomness

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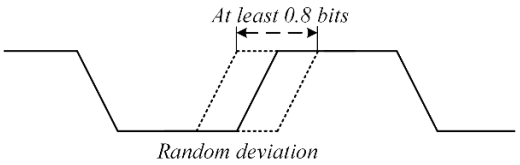

Fig. 1. Schematic diagram of jitter.

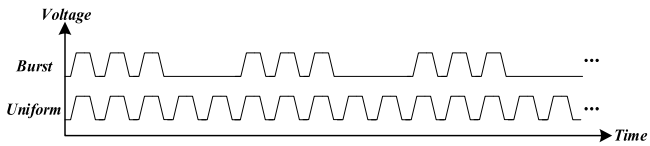

Fig. 2. Burst mode and uniform of self-timed rings (STRs).

in STRs and describes the relevant research. In section III, we theoretically prove why the proposed method improves the randomness. Section IV describes the designed experimental device. Section V provides randomness results and discusses them. Conclusions are drawn in section VI.

II. Related Research

A. Generation and Randomness of Jitter

There is no ideal digital signal in an actual circuit. As shown in Fig. 1, due to thermal noise, the rising and falling edges of the sampling waveform sometimes advance or lag, which is called jitter.

The "urn" model of Sunar *et al.* was pioneering work in the study of the randomness of jitter [20]. They pointed out that the jitter phenomenon approximately follows the normal distribution $N\left(\mu, \sigma^2\right)$, where μ is the position when the ideal edge changes and σ^2 is the variance. One jitter contains at least 0.8 bit of entropy, so the jitter width w has a minimum value of 1.3998 σ. If the jitter value is constrained to be less than or equal to half the oscillation period of RO $\left(T_{ro}/2\right)$, then the probability (entropy) of jitter can be expressed as

$$P_s\left\{\mathrm{w} \leq \frac{T_{ro}}{2}\right\} = \int_{\mu-\frac{w}{2}}^{\mu+\frac{w}{2}} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx. \tag{1}$$

B. Randomness of STRs

Each step in STRs consists of an inverter and a Muller C-element, which has a forward and reverse input (the input connecting the inverter). By adjusting the configuration of STRs during initialization, their oscillation behavior can be changed in the time domain. For future reference, we introduce the concepts of the bubble (B) and token (T). For each step of STRs, if the output and input of the step are the same, then the step is marked as B, and otherwise as T.

Anthony *et al.* pointed out that STRs have two modes of oscillation [21]: burst and uniform. As shown in Fig. 2, the time intervals of the oscillation waveform are not the same in burst mode, and they are the same in uniform mode. The designed STRs produce randomness in uniform mode.

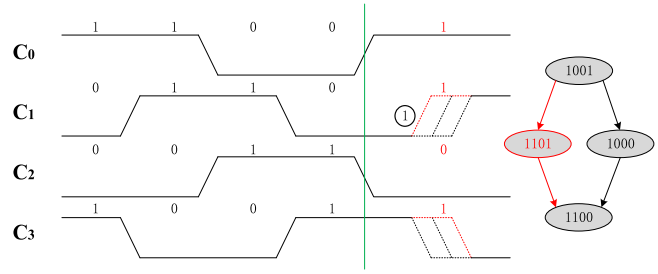

Fig. 3. Randomness in self-timed rings (STRs). The occurrence of jitter (right side of green line in Fig. 3) will cause transient instability during state transition. The dotted line in Fig. 3 represents the jitter width.

Liu *et al.* found in an experiment that in the uniform oscillation mode, assuming that the Charlie delay [22] of each stage of STRs is the same, the state transition diagram when the initial value is configured as "1001" (TBTB) will be as shown in Fig. 3, [17] and the state transition of the four output terminals C_0–C_3 of the four stages of STRs is 1001 → 1100 → 0110 → 0011 → 1001 (left side of green line).

In the process of state transition 1001 → 1100, in the jitter shown as red path 1 in Fig. 3, the intermediate state of "1101" will be introduced. That is, there are two possible results in the original opposite oscillation state (if the jitter direction is opposite to the legend, the unstable state of "1000" will be introduced), and the value only depends on the jitter direction. Since jitter follows a normal distribution, the unstable state is also normally distributed.

III. Influencing Factors and Improving Principles of Randomness

A. Influencing Factors of Randomness

The common TRNG sampling method is to directly sample the entropy source with a trigger, [11], [14], [17] and the probability that a sampling point appears at each position in the oscillation period is the same. It can be seen from the results in formula (1) that if the oscillation period of the periodic signal generated by STRs is T_{str}, then the probability density of the sampling point can be expressed as $1/T_{str}$, so that the sampling coverage (probability from sampling to jitter in one oscillation period) can be calculated as

$$P_{sc} = \frac{2P_s w}{T_{str}}. \tag{2}$$

where P_s is the probability of jitter occurrence, and w is the jitter width.

It can be seen from equation (2) that the sampling coverage of jitter in a period (P_{sc}) is very small because w is far less than T_{str}. For the above sampling method, even if the sampling frequency is increased so that multiple sampling can be carried out in a cycle, jitter extracted from a single sampling is still a small-probability event. Therefore, the low coverage of jitter directly affects the quality of the sampling rate and randomness.

The influence of process deviation on randomness cannot be ignored. TRNG is a technology to amplify unstable factors (such as circuit jitter) under small delays to generate random

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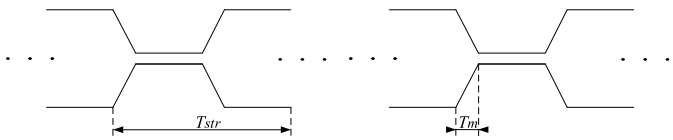

Fig. 4. The length of the sampling interval is shortened.

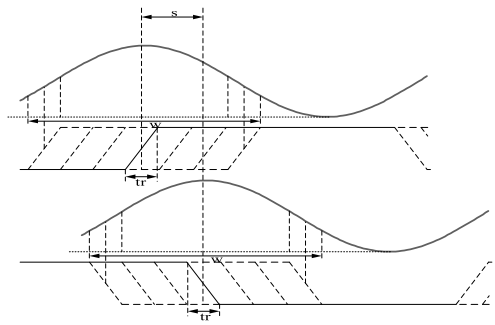

Fig. 5. Jitter model of any stage in self-timed rings (STRs).

numbers. But the process deviation can also bring variant delay, so that the unstable dalay by jitter may be covered. In order to reduce the impact of this problem, Anandakumar *et al.* designed a configurable logic block (CLB) to fine-tune the delay to reduce the influence of process deviation [14]. However, this will inevitably introduce cumbersome placing and routing, which directly limits the portability of TRNG.

B. Principle of Improving Randomness

In this paper, the probability density of the sampling point is increased by $1/T_{str}$ to $1/T_m$ to improve the sampling coverage of jitter. The edge state of the STRs is detected by a Muller C-element. By configuring the STRs uniformly, two completely opposite oscillation states are sampled. As shown in Fig. 4, since the Muller C-element is in the hold state when the input is opposite, it will sample only when jitter occurs. This can effectively shorten the sampling time and improve coverage.

The separation time s (the absolute difference between the arrival time of two input terminals of any stage in STRs) will be introduced in the process deviation, which will cause the output result to be biased. We discuss how to improve the output deviation through the designed post-processing structure in section IV.B.

Fig. 5 shows the jitter model of two inputs of any stage in STRs, where s is the separation time, w is the range of jitter, and tr is the rise time of the edge. (To facilitate calculation, the rise time is ignored in this model.) The upper part is the edge jitter distribution model. To facilitate analysis and observation, we enlarge the jitter width in Fig. 5; the lower part is the oscillation waveform of the STRs.

Because of the separation time s ($s < w$), the randomness of the two inputs of STRs will change, but at each stage, the probability of jitter is still

$$g(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}. \tag{3}$$

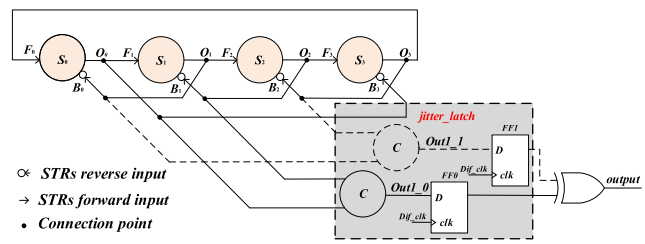

Fig. 6. Proposed connection between entropy source and jitter-latch structure.

After the influence of separation time is introduced, the simplified model of entropy is

$$P_m\left\{w \le \frac{T_{str}}{2}\right\} = 2\int_{\mu-\frac{w}{2}+s}^{\mu+\frac{w}{2}} g(x)\,dx. \tag{4}$$

For the designed jitter-latch structure, since the sampling point will appear only when jitter occurs, the probability of the sampling point $t \in \left[-\frac{w}{2}, \frac{w}{2}\right]$ appearing in the waveform is also subject to the normal distribution $g(x)$. The interval length is weighted to obtain the mean value, and $\mu = 0$, so the mean value of the sampling time length is

$$\begin{aligned} T_m &= \int_{-\frac{w}{2}}^{\frac{w}{2}} |tg(t)|\,dt - s = \int_0^{\frac{w}{2}} \left|\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}} dx^2\right| - s \\ &= \frac{\sqrt{2}\sigma}{\pi} e^{-\frac{w^2}{8\sigma^2}} - s \ll T_{str}. \end{aligned} \tag{5}$$

After the entropy P_m is substituted in equation (5), considering the sum of separation time and interval width, the sampling coverage rate of one sampling in this model is

$$P_{mc} = \frac{2P_m\,(\text{w-s})}{\frac{\sqrt{2}\sigma}{\pi} e^{-\frac{w^2}{8\sigma^2}} - s}. \tag{6}$$

By changing the condition of formula (6) to the nth sampling, the sampling coverage of jitter in the nth sampling can be obtained as

$$P_{nmc} = 1 - \prod_{i=1}^{n}\left(1 - \frac{2P_m\,(w_i - s)}{\frac{\sqrt{2}\sigma}{\pi} e^{-\frac{w^2}{8\sigma^2}} + s.}\right). \tag{7}$$

It can be seen from (5) and (6) that since $T_m \ll T_{str}$, P_m/P_s, and (w-s)$/w$ are not infinitesimal, the sampling coverage $P_{mc} > P_{sc}$. At the same time, equation (7) shows that multiple sampling can further improve the sampling coverage. Combining the two conclusions, we introduce the jitter-latch structure to replace the traditional trigger sampling and improve the sampling coverage of jitter by multi-bit output XOR.

IV. Design of TRNG

A. Design of Jitter-Latch Structure

Fig. 6 shows the schematic diagram of the jitter-latch structure including STRs. S_0–S_3 are four stages of STRs. The structure of the jitter-latch is shown in the gray area in Fig. 6. It is composed of a Muller C-element and a D flip-flop. Its two input terminals are connected to the output of K + n/2 and Kth of the n-th (n is an even number) STR.

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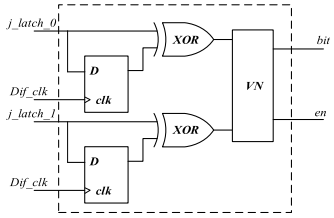

Fig. 7. Internal diagram of post-processing structure.

The D flip-flop is used to store the sample value of the C unit, which uses the differential clock *Dif_clk* (400 MHz), and finally processes the results of all jitter-latches into 1-bit data through an XOR. Combining the principle of randomness in section II.B, using the Muller C-element detects the oscillation edges of two STRs at the output end and latch the outliers caused by jitter, and then outputs them as random numbers. When the oscillation waveforms are completely opposite, the jitter-latch will always keep the original output unchanged, and when the oscillating waveform has a short uncertain state, this small uncertain state of the edge can be captured and expanded to a random waveform.

Fig. 8 (a) shows the output result when no jitter occurs. In this case, the nth order of STR and the corresponding n $\pm$ 8th order are always opposite, and the output value of the jitter-latch structure is always zero. Fig. 8 (b) shows the output result when jitter occurs. The jitter in the yellow circle makes the STR oscillation state appear as a short intermediate state (as shown in the red circle). At this time, the jitter-latch structure has latched the intermediate state and waited. When the rising edge of the clock shown in the blue circle arrives, the flip-flop is sampled to an intermediate state and converted to a random output. This can be changed from the value of the jitter-latch structure in the first row in Fig. 8 (b).

B. Design of Post-Processing Structure

The jitter-latch structure will be affected by process deviation, separation time, and the transition time of the Muller C-element. The fixed offset will always make the sampling results of the jitter-latch structure tend to be either "1" or "0." To filter the offset, we introduce a comparison loop and post-processing structure, using the von Neumann correction method to process data into random bit streams. Its structure is shown in Fig. 7. The input is the sampling result of the jitter-latch structure, containing a D flip-flop, an XOR gate, and judgment logic (*VN* in Fig. 7), and the output includes a random bit output terminal (*bit*) and a trigger enable output terminal (*en*).

The post-processing structure filters the offset by buffering the input terminal through the flip-flop for a period, and then XOR with its own secondary state; when no jitter occurs, the *XOR* result in the selector unit will always remain the same. Then the output of the *VN* structure will be judged according to the following situations:

(1) When the upper and lower *XOR* results have the same result, there are two possible situations: either the result caused by the offset phenomenon is the same or two STRs have the same jitter direction.

(2) When the upper and lower *XOR* results are different, there is only one possibility: at least one of the two STRs has jitter.

Since the *XOR* results contain the offset factor, the same results can be filtered to ensure that the randomness of TRNG output is generated by jitter. When the *XOR* results are the same, the enable output terminal does not send the enable signal, and the trigger group does not work; when the exclusive or result is different, the enable output terminal sends the enable signal, the register group works, and the output terminal transmits the corresponding output to the trigger. The output of the post-processing structure is set to "0" when the XOR results of the respective upper and lower input terminals are "0, 1," and "1" when "1, 0," to generate a stable and unbiased true random number.

C. Experimental Setup of Proposed TRNG

Cherkaoui *et al.* pointed out that increasing the order of an STR can improve stability, [19] comprehensive resource overhead, and other factors. Two 16-order STRs are selected as entropy source circuits to generate randomness. Winstanley *et al.* showed that STRs can maintain a constant phase difference between different outputs [21]. To ensure that the oscillation state of the STR can meet the experimental requirements of this article, we only place two tokens (T) at the beginning and the middle of the STR ring, and fill the rest with bubbles (B). Through this configuration, the *n*th order of the 16-stage STR and the corresponding $n \pm 8$th order can be made. The waveform is always the opposite, and this configuration cannot only minimize the switching activity to reduce power consumption but can give the jitter-latch of each stage a long enough setup and hold time to prevent incorrect sampling of jitter data. Since the STR ring contains eight jitter-latches that can independently output random bit streams, the proposed design still has a considerable throughput rate in this configuration.

The proposed TRNG was implemented on a Xilinx Virtex-6 FPGA and Xilinx Spartan-6 FPGA. Fig. 9 is the block diagram of each TRNG module. The *JL-TRNG* module is the main part of the proposed design, whose function is to generate truly random numbers. It has three inputs: clk provides a global clock signal, Dif_clk provides a high-speed sampling clock signal, and S initializes the STR configuration. The *Microblaze* module is the FPGA's resource, whose function is to initialize and configure the entropy source structure of TRNG and send output random numbers to the *PC*. The register module stores random numbers. *DCM* is a digital clock management module that provides clock signals to other modules.

The proposed TRNG contains two jitter-latch structures (*j_latch_0* and *j_latch_1*), each consisting of a 16-stage STR and eight jitter-latches. Each stage of the STR is a five-input LUT whose input terminals consist of a forward input, reverse input, clear, set number, and feedback terminal (used to realize the transfer of the secondary state). Each jitter-latch is a three-input LUT, including two input terminals and a feedback terminal. The jitter-latch is followed by the post-processing structure (*Selector* in Fig. 9) and connected with the last-stage

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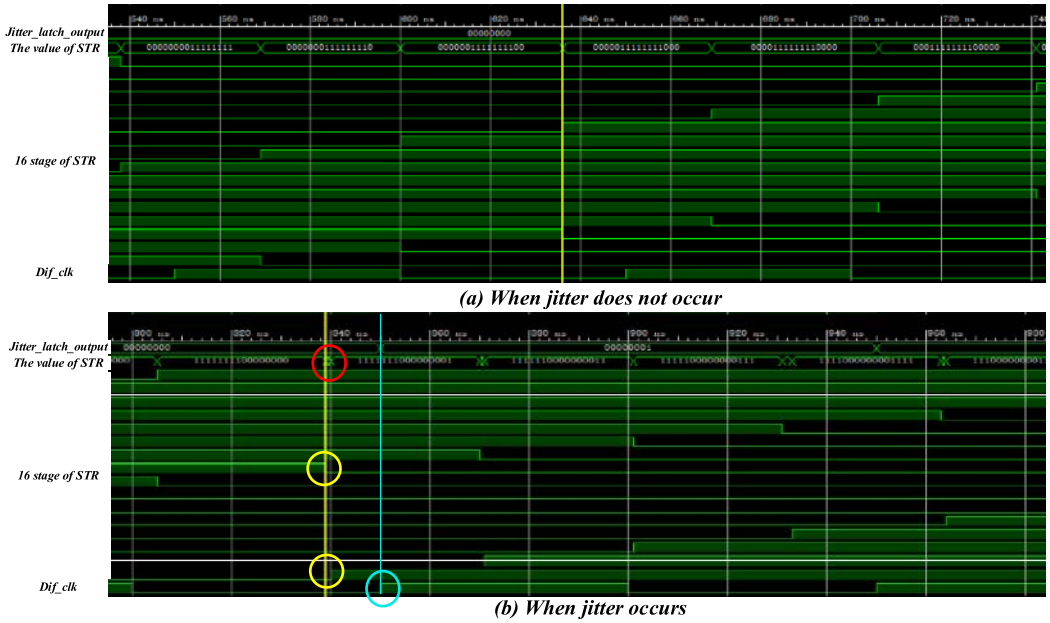

Fig. 8. Simulation result of output of jitter-latch structure. The first line of the waveform is the output of the jitter-latch, the second is the STR state transition, and the third to 18th lines are the 16-level STR. The last line of the waveform is the sampling clock Dif_clk.

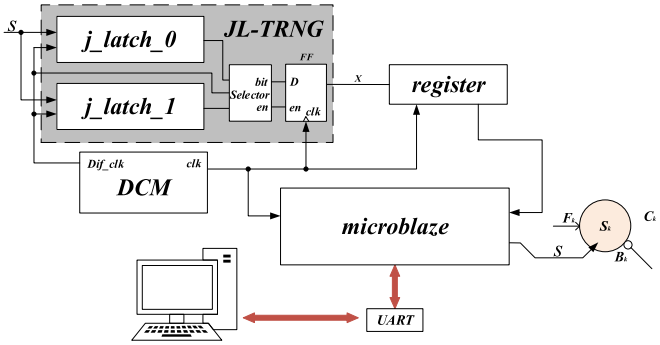

Fig. 9. Block diagram of TRNG modules.

flip-flop. Finally, we temporarily store the random number generated in the register.

V. Experimental Results

A. Automatic Placing and Routing to Verify Portability

The portability reference [16] described in this paper refers to a design that only needs to adjust the compilation method without adjusting the circuit structure after changing the device type, underlying logic, or process technology. This requires that TRNG does not rely on the unique structure of FPGA, and does not introduce manual placement or routing in the routing process.

TRNG is a technology to amplify unstable factors (such as circuit jitter) under small delays to generate random numbers. However, due to the slight deviation in the process of the development board, there is a difference in the delay between the actual and designed circuits. Most experimental methods are affected by the delay difference in the process of placing the oscillation loop, which will cause correlation in the results. Therefore, manual placement or routing should be used to place the oscillatory loop to reduce the interference caused by the delay difference. This affects the portability of the structure, which makes it necessary to modify the code to adjust the wiring position of the circuit after replacing the equipment.

The proposed TRNG is placed in multiple FPGAs by automatic routing without adding additional wiring structure, which leads to incomplete symmetry of the circuit layout and the difference of entropy source circuit. Through repeated experiments on different FPGAs, the influence of process deviation on the output of the circuit is simulated, and the portability of the design is tested to the maximum extent.

Fig. 10 shows part of the circuit after automatic placement and wiring. Fig. 10 (a) and (b) show two routing graphs of jitter-latch structures, which are not completely symmetric random distribution patterns, while the corresponding post-processing structures are distributed nearby; the red line in Fig. 10 (c) indicates the two order routing patterns in str. It can be seen that even if two LUTs are placed in the same piece, there are routing differences. Therefore, this method can be used to check whether the design structure has a certain tolerance limit to the process deviation, i.e., whether it has portability.

B. Random Test

The randomness test evaluates the randomness of TRNG; the most important index is the correlation of the random number. The TRNG structure designed in this paper was implemented on Xilinx Virtex-6 FPGAs and Xilinx Spartan-6 FPGAs. Under different working conditions, a series of one million bits were generated continuously to test the randomness.

1) Restart Test: We tested the randomness by sampling the initial bits in the experiment many times under the same

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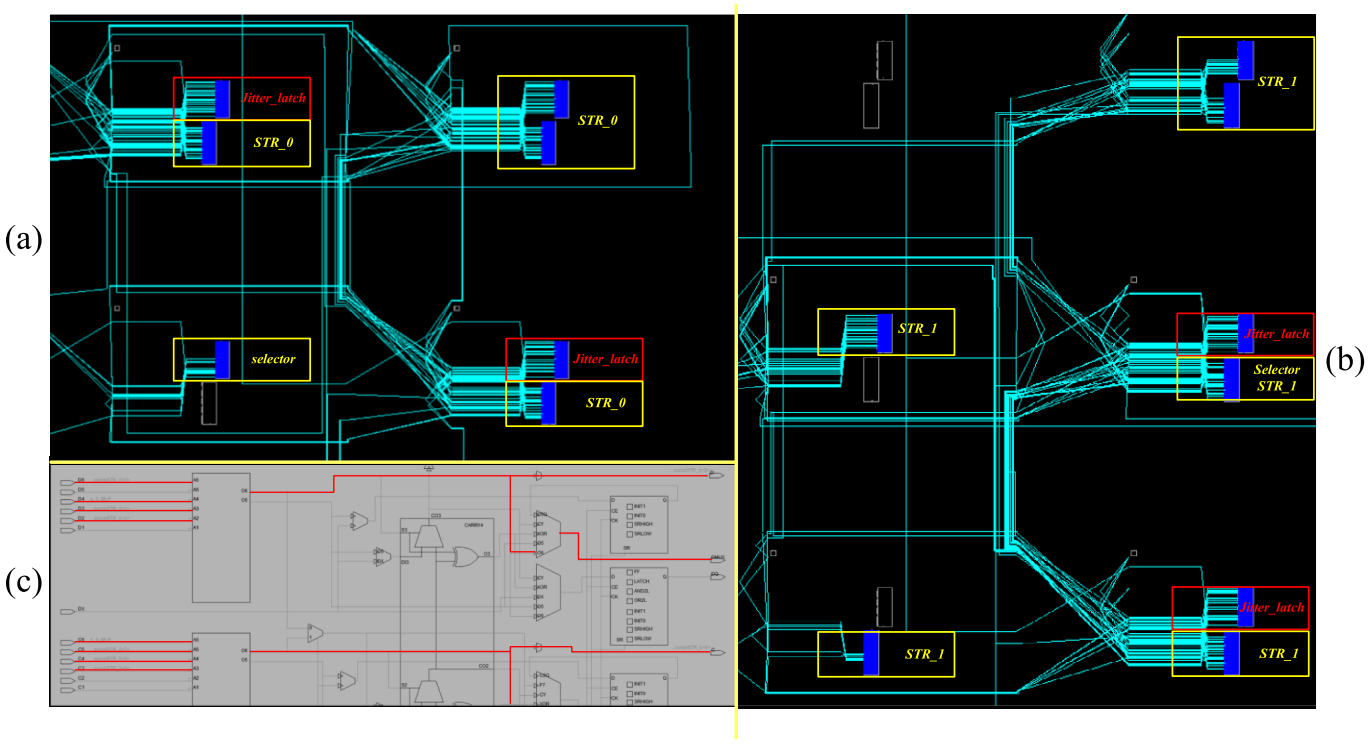

Fig. 10. Proposed TRNG under automatic placing and routing.

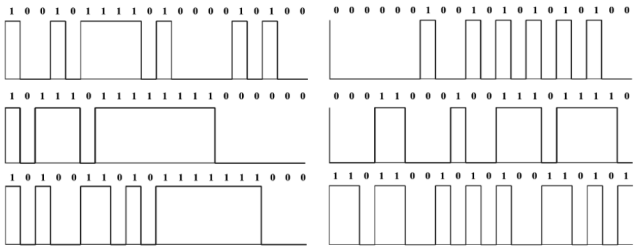

Fig. 11. Restart test results.

starting conditions. The data of the first 20 sampling bits after six restarts were observed and plotted, as shown in Fig. 11. For the pseudo-random sequence, when the random number generator starts repeatedly under the same starting conditions, the restart experiment will always produce a similar wave form. However, in our experimental results, it can be observed that the results of each restart were different, hence the proposed TRNG does not produce a pseudo-random sequence.

2) Deviation Test: The offset degree of a random number can be reflected by the proportion of "0" and "1" values at a certain data length. The ideal state is 50% each, because the offset degree will not only reduce the randomness but will make TRNG more vulnerable to attack [23]. We conducted the experiment on two Xilinx Virtex-6 FPGAs and two Xilinx Spartan-6 FPGAs, and carried out offset and autocorrelation tests on their generated random number sequences. The test results are shown in Fig. 12. On the left are four images generated by one million consecutive bits output by the proposed TRNG under standard working conditions (25°C, 1.0 V). In Fig. 12, the black and white pixels represent "1" and "0," respectively. Biased data will have too many black or white pixels in the bitstream image; we can see that the data distribution is uniform in the bitstream image. The offsets of the output sequences of the four FPGAs calculated by MATLAB are −0.0484%, 0.008%, 0.0672%, and −0.0948%, which shows that the output data have no offset.

3) Autocorrelation Test: We used the autocorrelation function of MATLAB to test the output of the designed TRNG, setting the lag value from 1 to 100, transforming the static model to a dynamic model, and analyzing the correlation coefficient between its output and 100 lag variables. According to the statistical index designed by Karl Pearson, when the correlation coefficient is less than 0.3, it can be considered irrelevant. The test results are shown on the right (a–d) in Fig. 12. The horizontal axis is the lag value, and the vertical axis is the autocorrelation coefficient. The correlation coefficient between the results and each lag value is close to 0, indicating no autocorrelation in the random sequence generated by the method in this paper.

4) NIST Test: The randomness of the output data of four FPGAs was further analyzed by the NIST SP 800-22 and NIST SP 800-90B tests. NIST SP 800-22 is used to evaluate the entropy of random strings, where the P-value is an indicator of the randomness of sequences, and "Prop." is the proportion of passing tests. When the P-value is greater than or equal to 0.01, the result has good randomness. NIST SP 800-90B was the latest test method proposed for testing the entropy source of TRNGs, where p-max is the probability of observing the most common sample, and h-min is the lowest possible entropy estimate.

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TABLE I
RANDOMNESS TEST OF VIRTEX-6 FPGAS AND SPARTAN-6 FPGAS

NIST SP 800-22	Virtex-6 FPGAs				Spartan-6 FPGAs			
Randomness Test	Chip 1		Chip 2		Chip 1		Chip 2	
	P-value	Prop.	P-value	Prop.	P-value	Prop.	P-value	Prop.
Approximate Entropy	0.954872	100%	0.793010	99%	0.276647	96%	0.500934	99%
Block Frequency	0.455937	98%	0.891893	100%	0.930811	100%	0.701057	99%
Cumulative Sums 1	0.946308	100%	0.986478	100%	0.373353	97%	0.553576	95%
Cumulative Sums 2	0.289667	99%	0.958164	100%	0.416040	98%	0.514728	97%
Fast Fourier Transform (FFT)	0.218821	99%	0.462869	99%	0.930626	100%	0.551917	99%
Frequency	0.834308	100%	0.936237	100%	0.298159	96%	0.472757	97%
Linear Complexity	0.304126	99%	0.261078	98%	0.431020	97%	0.261680	97%
Longest Run	0.897763	99%	0.034182	99%	0.757726	97%	0.710117	99%
Overlapping Template	0.964295	100%	0.336406	98%	0.809646	98%	0.709461	99%
Rank	0.012650	100%	0.240260	99%	0.917662	100%	0.570795	100%
Runs	0.779188	98%	0.702466	99%	0.928084	100%	0.926288	100%
Serial 1	0.953849	100%	0.440482	100%	0.409671	99%	0.090936	99%
Serial 2	0.368099	98%	0.179214	100%	0.620451	97%	0.043745	100%
Universal	0.818724	100%	0.759952	99%	0.573989	98%	0.462042	99%

NIST SP 800-90B	Virtex-6 FPGAs				Spartan-6 FPGAs			
p-max	Chip 1		Chip 2		Chip 1		Chip 2	
MCV	0.501370	0.502765	0.502151	0.502206	0.501887	0.502873	0.502495	0.501860
Collision	0.515625	0.534180	0.531250	0.527344	0.535156	0.539062	0.533203	0.537109
Markov	4.28E-39	4.48E-39	3.95E-39	4.78E-39	4.12E-39	4.46E-39	4.40E-39	4.26E-39
Compression	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
t-Tuple	0.528461	0.531208	0.521252	0.519794	0.521472	0.526041	0.520882	0.520695
LRS	0.499980	0.507602	0.502247	0.500360	0.512459	0.502081	0.507586	0.502358
Multi-MCW	0.502085	0.502178	0.501873	0.502283	0.501139	0.501723	0.502321	0.503449
Lag	0.501684	0.502032	0.502545	0.501206	0.501700	0.501222	0.502114	0.502200
Multi-MMC	0.501378	0.500961	0.502855	0.502536	0.500844	0.502848	0.501509	0.501994
LZ78Y	0.501924	0.502180	0.501156	0.501902	0.501225	0.502164	0.502243	0.501579
h-min	Chip 1		Chip 2		Chip 1		Chip 2	
MCV	0.996052	0.992044	0.993806	0.993649	0.994565	0.991733	0.992819	0.994644
Collision	0.955606	0.904603	0.912537	0.923184	0.901968	0.891476	0.907243	0.896712
Markov	0.995763	0.995246	0.996666	0.994527	0.996181	0.995292	0.995462	0.995828
Compression	1	1	1	1	1	1	1	1
t-Tuple	0.920131	0.912651	0.939946	0.943988	0.939338	0.926752	0.940971	0.941488
LRS	1.000060	0.978230	0.993531	0.998963	0.964491	0.994008	0.978276	0.993212
Multi-MCW	0.993998	0.993730	0.994606	0.993428	0.996716	0.995037	0.99332	0.990081
Lag	0.995151	0.994147	0.992674	0.996524	0.995104	0.996479	0.993913	0.993665
Multi-MMC	0.996029	0.997229	0.991784	0.992701	0.997567	0.991805	0.995653	0.994257
LZ78Y	0.994461	0.993725	0.996669	0.994522	0.996471	0.993769	0.993543	0.995451

TABLE II
TEMPERATURE AND VOLTAGE CONDITIONS OF TEST ITEMS

Case 1	Case 2	Case 3	Case 4	Case 5
0°C, 0.9 V	0°C, 1.0 V	0°C, 1.1 V	25°C, 0.9 V	25°C, 1.0 V
Case 6	Case 7	Case 8	Case 9	
25°C, 1.1 V	80°C, 0.9 V	80°C, 1.0 V	80°C, 1.1 V	

Our test results are shown in Table I. We output 400 groups of data with a length of a continuous one million bits from two Virtex-6 FPGAs and two Spartan-6 FPGAs. It can be seen from the results that all the random bitstreams generated by the proposed TRNG can pass all tests of randomness.

C. Process Verification Test (PVT)

As previously stated, changes in the manufacturing process and environment may affect the quality of a TRNG, whose good design will resist certain PVT changes. We evaluated the performance of a TRNG at different temperatures and voltages on a Xilinx Virtex-6 FPGA.

The NIST SP 800-90B test was selected for the PVT test. NIST SP 800-90B is more inclined than NIST SP 800-22 to test the results of a random number generator based on a

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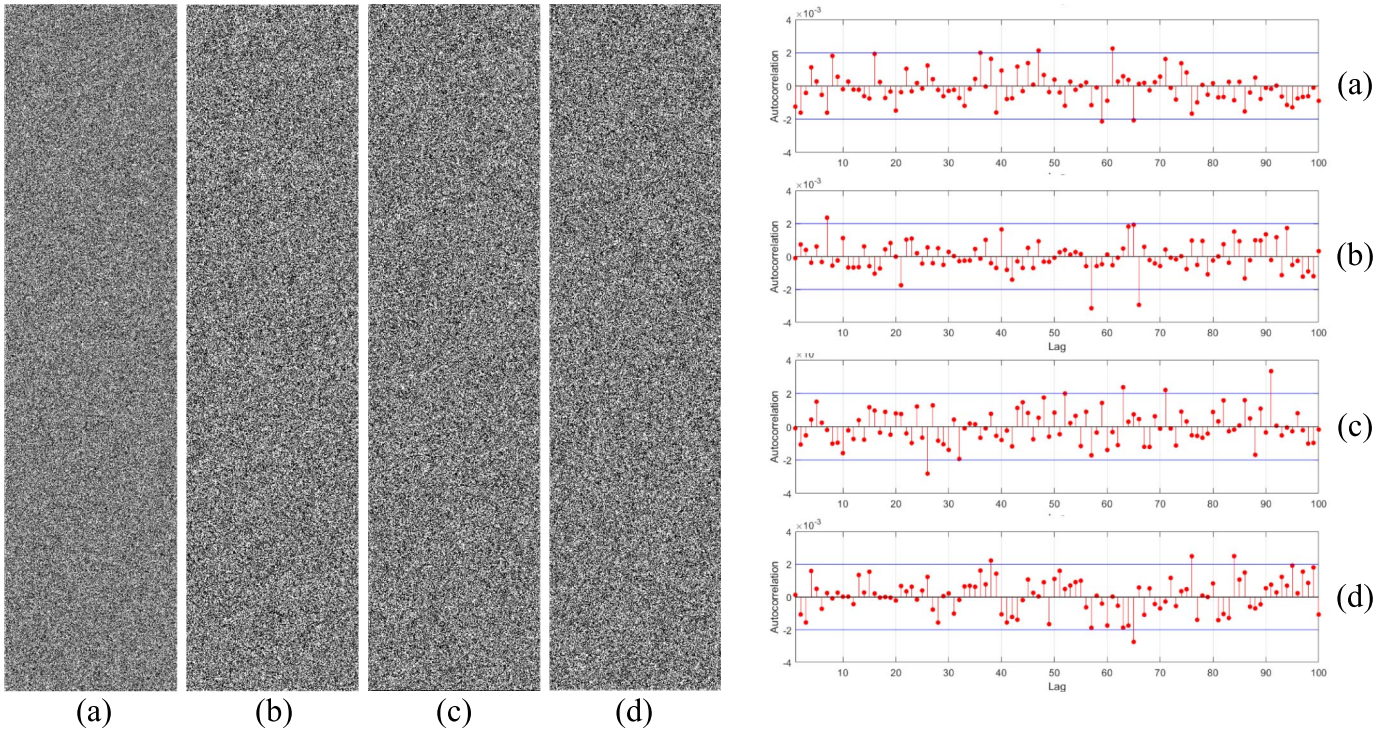

Fig. 12. Bitstream image (pixel size 500 × 2000) and autocorrelation in standard state.

TABLE III
RESULTS OF NIST SP 800-90B TEST

p-max	Case 1	Case 2	Case 3	Case 4	Case 5*	Case 6	Case 7	Case 8	Case 9
MCV	0.501624	0.501754	0.501357	0.501647	0.501544	0.501530	0.501328	0.501776	0.501929
Collision	0.533203	0.536133	0.527344	0.531250	0.533203	0.544922	0.517578	0.537109	0.531250
Markov	3.87e-39	3.89e-39	4.09e-39	3.78e-39	3.70e-39	4.52e-39	3.71e-39	3.73e-39	4.06e-39
Compression	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
t-Tuple	0.527557	0.526629	0.520025	0.520025	0.519390	0.526629	0.526629	0.526629	0.526629
LRS	0.510206	0.502551	0.535330	0.499923	0.502963	0.504700	0.504565	0.500817	0.515539
Multi-MCW	0.556641	0.500734	0.500305	0.500772	0.501380	0.501364	0.501312	0.500729	0.502093
Lag	0.501335	0.505859	0.500522	0.501556	0.500887	0.501446	0.501011	0.501329	0.501140
Multi-MMC	0.501389	0.500542	0.501471	0.519531	0.501393	0.500904	0.501073	0.502216	0.501048
LZ78Y	0.501367	0.500512	0.501480	0.500979	0.500834	0.501404	0.500233	0.501457	0.501270
h-min	**Case 1**	**Case 2**	**Case 3**	**Case 4**	**Case 5***	**Case 6**	**Case 7**	**Case 8**	**Case 9**
MCV	0.995322	0.994948	0.996090	0.995256	0.995552	0.995592	0.996173	0.994885	0.994445
Collision	0.907243	0.899338	0.923184	0.912537	0.907243	0.875879	0.950151	0.896712	0.912537
Markov	0.996886	0.996835	0.996259	0.997134	0.997399	0.995158	0.997371	0.997325	0.996345
Compression	1	1	1	1	1	1	1	1	1
t-Tuple	0.922601	0.925141	0.943346	0.943346	0.945111	0.925141	0.925141	0.925141	0.925141
LRS	0.970847	0.992657	0.901500	1.000220	0.991475	0.986503	0.986889	0.997645	0.955848
Multi-MCW	0.845182	0.997885	0.999119	0.997776	0.996025	0.996071	0.996221	0.997900	0.993975
Lag	0.996155	0.983192	0.998493	0.995519	0.997444	0.995835	0.997087	0.996172	0.996716
Multi-MMC	0.995998	0.998437	0.995762	0.944718	0.995986	0.997394	0.996907	0.993620	0.996979
LZ78Y	0.996062	0.998522	0.995737	0.997179	0.997597	0.995956	0.999326	0.995803	0.996342

*Case 5 is the random test result under the standard working condition (25°C, 1.0 V).

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TABLE IV
RESOURCE COST COMPARISON RESULTS OF RELATED TRNGS

Entropy source	Resource cost	Portability	Operating frequency	Average P-value	Throughput
ROs [11]	40 slices	No	/	/	14.3 Mbps
ROs [14]	526 LUTs and 177 DFFs	No	24 MHz	0.4517	6 Mbps
STRs [15]	320 LUTs and 320 DFFs	No	400 MHz	/	200 Mbps
STRs [16]	32 LUTs and 48 DFFs	Yes	1 MHz (50 MHz test failure)	0.2–0.8*	4 Mbps
STRs [17]	/	No	100 MHz and 400 MHz	0.5736	100 Mbps
This work	56 LUTs and 19 DFFs	Yes	100 MHz and 400 MHz	0.6285	100 Mbps

/ There is no specific data in the reference. * There is only the range of NIST test results in the reference

physical design [23]. The working conditions of test items in cases 1–9 are shown in Table II, and the test results are shown in Table III. We can see from the results that the output p-max and h-min of the propose3d TRNG tend to be stable under different PVT conditions.

D. Comparison With Other TRNGs Based on FPGA

The total logic power obtained from ISE power analysis results with the proposed TRNG is 1.15 mW. This structure is also compared with other TRNGs based on FPGAs in terms including resource cost, throughput, portability, and operating frequency. The results are shown in Table IV. The TRNGs proposed by all five references are pure digital TNRGs based on FPGAs, and all passed or partially passed the random test. In terms of resource overhead, the main structure of our proposed TRNG is composed of 56 LUTs and 19 DFFs, which are automatically generated by EDA tools. It realizes portability and high randomness, with a compromise between resource overhead and high throughput compared with the four references.

VI. CONCLUSION

A TRNG based on a jitter-latch structure is proposed. We used the oscillations of the STRs to generate randomness, and the jitter-latch structure to accurately extract the randomness, improve the sampling coverage of jitter, and increase the output entropy of the structure. A post-processing structure was proposed to filter out most of the deviation in the placing and routing process. The portability of the structure was verified on different FPGAs by automatically routing with the EDA tool. Experimental analysis showed that this method can effectively improve the sampling coverage of jitter and reduce the offset phenomenon in the sampling results. The designed TRNG has stable randomness and high throughput. It has a certain optimization ability for the offset caused by process deviation, which provides the possibility of a TRNG design based on FPGA that can be transplanted on an ASIC circuit.

REFERENCES

[1] M. Hilbert and P. Lopez, "The World's technological capacity to store, communicate, and compute information," *Science*, vol. 332, no. 6025, pp. 60–65, Apr. 2011.

[2] X. Wu and S. Li, "A new digital true random number generator based on delay chain feedback loop," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, Baltimore, MD, USA, May 2017, pp. 1–4.

[3] J.-L. Danger, S. Guilley, and P. Hoogvorst, "High speed true random number generator based on open loop structures in FPGAs," *Microelectron. J.*, vol. 40, no. 11, pp. 1650–1656, Nov. 2009.

[4] S. S. Saab, J. G. Hobeika, and I. E. Ouaiss, "A novel pseudorandom noise and band jammer generator using a composite sinusoidal function," *IEEE Trans. Signal Process.*, vol. 58, no. 2, pp. 535–543, Feb. 2010.

[5] R. Vaidyanathaswami and A. Thangaraj, "Robustness of physical layer security primitives against attacks on pseudorandom generators," *IEEE Trans. Commun.*, vol. 62, no. 3, pp. 1070–1079, Mar. 2014.

[6] O. Petura, U. Mureddu, N. Bochard, and V. Fischer, "Optimization of the PLL based TRNG design using the genetic algorithm," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, Baltimore, MD, USA, May 2017, pp. 1–4.

[7] E. N. Allini, O. Petura, V. Fischer, and F. Bernard, "Optimization of the PLL configuration in a PLL-based TRNG design," in *Proc. Design, Autom. Test Eur. Conf. Exhib. (DATE)*, Dresden, Germany, Mar. 2018, pp. 1265–1270.

[8] A. P. Johnson, R. S. Chakraborty, and D. Mukhopadyay, "An improved DCM-based tunable true random number generator for xilinx FPGA," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 64, no. 4, pp. 452–456, Apr. 2017.

[9] N. Fujieda, M. Takeda, and S. Ichikawa, "An analysis of DCM-based true random number generator," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 67, no. 6, pp. 1109–1113, Jun. 2020.

[10] P. Kohlbrenner and K. Gaj, "An embedded true random number generator for FPGAs," in *Proc. ACM/SIGDA 12th Int. Symp. Field Program. Gate Arrays (FPGA)*, Monterey, CA, USA, 2004, pp. 71–78.

[11] V. Rozic, B. Yang, W. Dehaene, and I. Verbauwhede, "Highly efficient entropy extraction for true random number generators on FPGAs," in *Proc. 52nd Annu. Design Autom. Conf. (DAC)*, San Francisco, CA, USA, Jun. 2015, pp. 1–6.

[12] D. Liu, Z. Liu, L. Li, and X. Zou, "A low-cost low-power ring oscillator-based truly random number generator for encryption on smart cards," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 63, no. 6, pp. 608–612, Jun. 2016.

[13] L. B. Carreira, P. Danielson, A. A. Rahimi, M. Luppe, and S. Gupta, "Low-latency reconfigurable entropy digital true random number generator with bias detection and correction," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 67, no. 5, pp. 1562–1575, May 2020.

[14] N. Nalla Anandakumar, S. K. Sanadhya, and M. S. Hashmi, "FPGA-based true random number generation using programmable delays in oscillator-rings," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 67, no. 3, pp. 570–574, Mar. 2020.

[15] A. Cherkaoui *et al.*, "A very high speed true random number generator with entropy assessment," in *Proc. CHES*, Berlin, Germany, 2013, pp. 179–196.

[16] H. Martin *et al.*, "A new TRNG based on coherent sampling with self-timed rings," *IEEE Trans. Ind. Informat.*, vol. 12, no. 1, pp. 91–100, Feb. 2016.

[17] Y. Liu, R. C. C. Cheung, and H. Wong, "A bias-bounded digital true random number generator architecture," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 64, no. 1, pp. 133–144, Jan. 2017.

[18] Y. Zhang, J. Jiang, Q. Wang, and N. Guan, "A self-timed ring based true random number generator on FPGA," in *Proc. 14th IEEE Int. Conf. Solid-State Integr. Circuit Technol. (ICSICT)*, Qingdao, China, Oct. 2018, pp. 1–3.

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[19] A. Cherkaoui, V. Fischer, A. Aubert, and L. Fesquet, "Comparison of self-timed ring and inverter ring oscillators as entropy sources in FPGAs," in *Proc. Design, Autom. Test Eur. Conf. Exhib. (DATE)*, Dresden, Germany, Mar. 2012, pp. 1325–1330.

[20] B. Sunar, W. J. Martin, and D. R. Stinson, "A provably secure true random number generator with built-in tolerance to active attacks," *IEEE Trans. Comput.*, vol. 56, no. 1, pp. 1070–1079, Mar. 2014.

[21] A. Winstanley and M. Greenstreet, "Temporal properties of self-timed rings," in *Proc. CHARME*, Berlin, Germany, 2001, pp. 140–154.

[22] J. C. Ebergen, S. Fairbanks, and I. E. Sutherland, "Predicting performance of micropipelines using Charlie diagrams," in *Proc. 4th Int. Symp. Adv. Res. Asynchronous Circuits Syst.*, San Deigo, CA, USA, 1998, pp. 238–246.

[23] S. Tao and E. Dubrova, "TVL-TRNG: Sub-microwatt true random number generator exploiting metastability in ternary valued latches," in *Proc. IEEE 47th Int. Symp. Multiple-Valued Log. (ISMVL)*, Novi Sad, Serbia, May 2017, pp. 130–135.

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